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Retinomorphic mid-wave infrared in-sensor processing engine: device-to-architecture co-design

Hemalatha Nagaraju

New Jersey Institute of Technology, hemank1911@gmail.com

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ABSTRACT

RETINOMORPHIC MID-WAVE INFRARED IN-SENSOR PROCESSING ENGINE: DEVICE-TO-ARCHITECTURE CO-DESIGN

**by
Hemalatha Nagaraju**

Conventional frame-based CMOS image sensors acquire full-frame pixel data at discrete time intervals, resulting in substantial spatial redundancy and loss of temporal information between frames. The repeated conversion and transfer of redundant pixel data increases bandwidth and power consumption in machine vision systems. Retinomorphic sensing architectures address these limitations by enabling programmable, analog-domain processing directly at the sensor interface. A compact behavioral model of the PbSe device is developed in HSPICE based on calibrated TCAD simulation data to capture gate-controlled photocurrent modulation under varying illumination and gate bias conditions. Error analysis is performed to quantify the deviation between TCAD-generated photocurrent characteristics and the circuit-level model to ensure simulation fidelity. To interface the crossbar array with CMOS circuitry, a mixed-signal front-end implemented in 180nm CMOS technology integrates a multi-bit DAC for gate programming, a high-value load resistor for passive current-to-voltage conversion, a transmission-gate sample-and-hold circuit, and a flash ADC for digitization. Simulation results validate end-to-end system operation and establish a circuit-level framework for programmable MWIR retinomorphic sensing, achieving over four to five orders of magnitude reduction in energy per frame compared to conventional cryogenically cooled MWIR imaging systems.

**RETINOMORPHIC MID-WAVE INFRARED IN-SENSOR
PROCESSING ENGINE:
DEVICE-TO-ARCHITECTURE CO-DESIGN**

by
Hemalatha Nagaraju

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APPROVAL PAGE

**RETINOMORPHIC MID-WAVE INFRARED IN-SENSOR
PROCESSING ENGINE:
DEVICE-TO-ARCHITECTURE CO-DESIGN**

Hemalatha Nagaraju

Dr. Shaahin Angizi, Thesis Advisor

Date

Assistant Professor of Electrical and Computer Engineering, NJIT

Dr. Dong-Kyun Ko, Committee Member

Date

Associate Professor of Electrical and Computer Engineering, NJIT

Dr. Anirudh Sridhar, Committee Member

Date

Assistant Professor of Electrical and Computer Engineering, NJIT

BIOGRAPHICAL SKETCH

Author: Hemalatha Nagaraju
Degree: Master of Science
Date: May 2026
Date of Birth: August 07, 2000
Place of Birth: Kunigal, Karnataka, India

Undergraduate and Graduate Education:

- Master of Science in Computer Engineering,
New Jersey Institute of Technology, Newark, NJ, May 2026
- Bachelor of Technology in Electronics and Communication Engineering,
Adichunchanagiri Institute of Technology, Chikkamagaluru, Karnataka, July 2022

Major: Computer Engineering

What is done with love is done well.

V. Van Gosh

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CHAPTER 1

INTRODUCTION

Imaging systems are essential elements of contemporary machine vision, supporting applications ranging from autonomous navigation to environmental monitoring. As computational capabilities continue to scale, the volume of visual data produced at the sensor level has grown significantly, resulting in greater demands on bandwidth, energy consumption, and real-time processing efficiency. Conventional imaging architectures separate sensing from processing, acquiring complete image frames that are subsequently digitized and analyzed [1]. Although this approach offers flexibility, it often leads to acquisition and transfer of data that may not be relevant for downstream tasks [2]. This limitation has prompted investigation into alternative sensor architectures that perform localized signal transformation prior to full frame readout. Retinomorphing sensing methods seek to replicate features of biological vision by enabling programmable modulation of photocurrent within the sensor array [3, 4]. Extending these capabilities to the mid-wave infrared (MWIR) spectral range facilitates passive imaging and operation in low-visibility conditions [5]. Achieving this functionality necessitates precise circuit-level integration to enable programmable biasing, photocurrent control, and mixed-signal conversion within a scalable array. This thesis examines the circuit-level implementation of a programmable MWIR retinomorphing sensing architecture.

1.1 Evolution and Limitation of Conventional Image Sensing Architectures

Solid-state image sensors utilizing complementary metal-oxide-semiconductor (CMOS) technology have become the primary platform for machine vision systems [6]. The integration of photodetectors, column-level readout circuits, and on-chip analog-

to-digital converters (ADCs) enables compact, cost-effective imaging solutions for diverse applications, including consumer electronics and industrial automation [7]. In conventional architectures, pixel arrays acquire complete image frames at discrete intervals, and the accumulated charge or voltage at each pixel is sequentially read, digitized, and transmitted to subsequent processing units.

While the frame-based acquisition model offers architectural flexibility, it also creates inefficiencies [2, 8]. Capturing full frames regardless of scene changes leads to spatial redundancy in static areas. Additionally, temporal information between frames is not explicitly retained, which limits responsiveness in high-speed or event-driven settings. Repeated conversion and transmission of pixel data increases bandwidth requirements and energy consumption, especially in edge-deployed systems where power and latency are critical [9].

1.2 Retinomorphic and In-Sensor Computing Paradigm

1.2.1 Biological and Event-Driven Sensing

Conventional image sensors capture visual information in discrete frames, generating sequences of complete pixel arrays for subsequent digital processing. In contrast, biological vision systems function without rigid frame boundaries. The human retina continuously senses and preprocesses visual input, transmitting compressed representations to higher visual centers. Photoreceptors convert incident light into electrical signals, while interneuron layers such as bipolar and ganglion cells modulate spatial and temporal responses to emphasize contrast and motion and suppress redundancy. This structure enables efficient encoding of dynamic scenes without the need to transmit full-resolution data at uniform intervals [2, 10, 11].

Inspired by biological vision, event-driven image sensors were developed to eliminate the need for continuous full-frame readout. Instead of outputting complete images at fixed intervals, event-based sensors report changes in illumination at

individual pixels, thereby avoiding periodic readout of the entire array [2]. This method reduces redundant data transfer and decreases bandwidth and latency compared to conventional frame-based imaging systems. Although event-based sensors primarily function in the visible spectrum and focus on temporal contrast detection, they demonstrate a key architectural principle: placing sensing and filtering elements closer to the pixel reduces redundant data transfer and improves efficiency [9, 12, 13].

1.2.2 Analog-Domain Processing and Crossbar Architectures

In-Sensor computing architectures extend beyond event-driven sensing by performing limited signal transformation directly within the sensor array prior to full digitization. Analog-domain processing reduces the necessity to convert and transmit every pixel value, thereby alleviating analog-to-digital converter (ADC) bandwidth and memory overhead in edge-vision systems [14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24]. Crossbar structures provide an effective platform for integrating in-sensor computing. When sensor elements are organized in a matrix configuration, photocurrents from individual nodes are summed along shared lines, which enables parallel accumulation according to Kirchhoff’s current law. This characteristic has been utilized in previous studies to implement compact filtering or weighted summation directly at the array level [25, 26].

Rather than relying on memristive behavior, the sensor response can be programmed through electrostatic modulation of the photodetector [12]. This approach enables electrical tuning of the photocurrent, facilitating multi-level analog outputs and compatibility with standard CMOS mixed-signal interfaces, including bias generation, low-current measurement, sampling, and digitization. The circuit-level details constitute the primary focus of this thesis and will be addressed in subsequent chapters.

1.3 Current State of the Field

Recent retinomorphic machine vision architectures are broadly classified into near-sensor and in-sensor processing approaches. Near-sensor designs exploit CMOS technology scaling to place low-power processing circuits adjacent to the sensor array [27, 28, 29, 30, 31], though the physical gap between sensing and computation still introduces latency and energy overhead [32, 33, 34, 35]. In-sensor processing eliminates this separation by embedding computation directly at the pixel level. Photoactive memristive devices have demonstrated sensor-memory-processor co-integration [36, 37], while synaptic phototransistors based on organic heterojunctions [38] and 2D ferroelectric materials [39] have shown photoadaptive behavior compatible with neural network inference.

Advances in 2D semiconductor optoelectronics have produced a broad range of in-sensor computing devices. Two-terminal structures based on $\text{Mo}_x\text{Re}_{(1-x)}\text{S}_2$ [40] and MoS_2 [41] achieve multicolor photoresponse through charge trapping and ion migration. Gate-tunable three-terminal devices, including $\text{WSe}_2/\text{h-BN}/\text{Al}_2\text{O}_3$ [3], WSe_2/GaN floating-gate transistors [42], $\text{PdSe}_2/\text{MoTe}_2$ heterojunctions [43], $\text{MoS}_2/\text{P(VDF-TrFE)}$ FETs [44], and $\alpha\text{-In}_2\text{Se}_3$ ferroelectric FETs [45] — enable programmable broadband detection. Multiterminal architectures further extend reconfigurability through split-gate ferroelectric homojunctions [46] and floating-gate charge trapping [47]. Despite this progress, all existing platforms are limited to visible wavelengths, motivating exploration of PbSe as an uncooled MWIR-active material [48, 49] for infrared retinomorphic sensing.

Wu et al. [50] demonstrated the first reconfigurable black phosphorus photodiode enabling in-sensor computing from visible to mid-infrared wavelengths (633–3324 nm). Gate-controlled electrostatic doping produces programmable bipolar photocurrent that maps directly to convolution kernel weights, achieving 96% accuracy on handwritten digit recognition. This work establishes the feasibility of mid-infrared

in-sensor computing but relies on black phosphorus, which lacks the low-cost fabrication compatibility of PbSe.

CMOS-based near-sensor [51, 52, 53] and in-sensor architectures [54, 55, 56, 57] have addressed visible-range feature extraction including Haar-like filtering and local binary patterns. However, these designs face thermal degradation and accuracy loss at high throughput [55, 58]. Advanced implementations such as 3D-stacked column-parallel ADC arrays [59], binarized neural network accelerators [60], and analog convolution engines [61] reveal an inherent trade-off between energy efficiency and inference accuracy that motivates the proposed MWIR retinomorphic architecture.

1.4 MWIR Sensing and Field-Effect Gated PbSe Devices

Mid-wave infrared (MWIR, 3–5 μm) imaging enables passive detection of thermal radiation emitted by objects, facilitating operation without external illumination. In contrast to visible-spectrum imaging, which depends on reflected light, MWIR sensing captures temperature-dependent radiative signatures [62, 5]. This capability is particularly effective in low-visibility environments such as smoke, haze, and fog, making MWIR imaging well-suited for environmental monitoring, industrial inspection, surveillance, and autonomous navigation, where resilience to variations in lighting and obscurants is essential. The ongoing demand for compact and energy-efficient infrared systems drives the investigation of device architectures that incorporate retinomorphic and in-sensor processing concepts within this spectral range.

To achieve retinomorphic functionality in the MWIR regime, a field-effect gated PbSe photodiode architecture employs gate biasing to electrically modulate the device response. Varying the gate voltage enables precise tuning of the photocurrent generated under MWIR illumination, both in magnitude and polarity. Gate-controlled programmability of the photocurrent enables multiple response

states, such as positive, negative, and near-zero output levels[12]. This behavior establishes a practical foundation for implementing analog-domain preprocessing directly at the sensor interface.

The feasibility and electrical characteristics of the programmable PbSe device were validated using TCAD simulations developed by Prof. Dong-Kyun Ko and his research group as part of a joint research effort. These simulations confirm the device-level operating principle and characterize its programmable photocurrent behavior under varying gate bias and illumination conditions.

1.5 Motivation

The programmable PbSe photodiode developed in the joint research effort offers a gate-controlled MWIR sensing platform with tunable photocurrent response. Realizing the full potential of this device requires a compatible circuit-level framework that can bias, sense, and digitize its programmable output. Integration with mixed-signal circuitry allows controlled gate programming, stable current-to-voltage conversion, and multi-bit quantization of the modulated photocurrent signal.

To support reliable circuit implementation, accurate compact model calibrated to TCAD simulation data is needed for HSPICE-based verification and system-level design. This allows the behavior of the programmable device to be incorporated into a structured analog front-end architecture. This thesis aims to develop a CMOS-compatible mixed-signal interface for the programmable MWIR PbSe device, with emphasis on scalable integration in crossbar-based sensing arrays. The resulting circuit framework provides a foundation for structured biasing, sampling, and digital conversion suitable for array-level retinomorphic implementations.

1.6 Thesis Organization

This thesis examines circuit-level modeling and mixed-signal integration of a field-effect gated PbSe photodiode for mid-wave infrared (MWIR) retinomorphic sensing. A TCAD-calibrated compact device model is developed and incorporated with a CMOS-compatible analog front-end and digitization framework for programming photocurrent operation within a crossbar architecture. The structure of the thesis is presented as follows:

Chapter 2: Compact Modeling of the Field-Effect Gated PbSe Photodiode: This chapter presents a TCAD-calibrated compact behavioral model for the PbSe photodiode. Device-level simulation data provided by the collaborating semiconductor research group are analyzed to develop an HSPICE-compatible model for gate-controlled photocurrent modulation. The modeling framework captures bias-dependent polarity transition, multi-level programmability, and illumination scaling, thereby establishing the device-level basis for subsequent circuit integration.

Chapter 3: Programmable Biasing and Crossbar Architecture: This chapter describes the programmable biasing strategy and crossbar configuration employed to operate the PbSe device array. A multi-level digital-to-analog conversion scheme is developed to generate structured gate voltages throughout the array. Crossbar-level current behavior and scalability are analyzed to establish a circuit-operable sensing framework.

Chapter 4: Analog Front-End and Signal Conditioning: This chapter develops a current-to-voltage conversion methodology utilizing a high-value load resistor and analyzes the resulting voltage swing characteristics. A transmission-gate sample-and-hold circuit is implemented to stabilize the analog signal before quantization. Signal integrity and dynamic response are evaluated within the mixed-signal interface.

Chapter 5: Multi-Bit Flash Analog-to-Digital Converter Design: This chapter details the design and simulation of a flash analog-to-digital converter (ADC)

implemented in 180 nm CMOS technology. The comparator architecture, reference ladder design, encoding logic, and quantization accuracy are analyzed to validate reliable digitization of the programmable photocurrent signal.

Chapter 6: System-Level Integration and Validation: This chapter integrates the compact device model, programmable biasing network, analog front-end, and ADC within a unified HSPICE simulation environment. Array-level operation and end-to-end signal flow are demonstrated to validate the feasibility of the proposed mixed-signal retinomorphic sensing framework.

Chapter 7: This chapter summarizes the technical contributions of this thesis, including the TCAD-calibrated compact model, the 4×4 PbSe crossbar array with programmable R-2R DAC biasing, the analog signal conditioning chain, and the 4-bit flash ADC. System-level integration results and architecture-level scalability analysis within the RISPE framework are reviewed.

CHAPTER 2

COMPACT MODELING OF PbSe PHOTODIODE

The previous chapter presented the device concept and circuit-level objectives of the programmable PbSe photodiode within a retinomorphic sensing framework. This chapter focuses on the device-level behavior that enables programmable photocurrent modulation and introduces a compact modeling methodology suitable for circuit integration. The discussion begins with the operating principle of the field-effect gated PbSe architecture, highlighting the significance of electrostatic gate control in forming reconfigurable junction profiles. Subsequent sections outline the TCAD simulation framework used to generate device characteristics and analyze bias- and illumination-dependent photocurrent behavior from the dataset. Based on this analysis, a TCAD-calibrated behavioral compact model is developed and implemented in HSPICE to facilitate mixed-signal co-simulation. This structured approach establishes the device-level foundation necessary for scalable crossbar and circuit-level integration.

2.1 Device Operation

The programmable PbSe photodiode examined in this study utilizes a field-effect gated architecture that enables electrostatic modulation of carrier distribution within a sensitized p-type PbSe photoconductive layer. The device structure includes independently biased n-gate and p-gate control terminals, as well as source and drain contacts for current collection. In contrast to conventionally doped photodiodes with fixed junction profiles, the electrostatic bias applied to the control gates dynamically modifies the internal electric-field distribution, enabling reconfigurable formation of p+-p-n or n-p-p+ junction configurations[49, 48]. The cross-sectional structure of

the device, showing the PbSe absorber layer, gate oxide, and independently biased gate terminals, is illustrated in Figure 2.1.

During operation, the source terminal is held at ground potential, and the drain current is measured as the electrical output. Differential gate biasing establishes controlled band tilting across the PbSe layer, which determines the drift and collection of photogenerated carriers under mid-wave infrared (MWIR) illumination. Applying opposite polarities to the two control gates results in bidirectional photocurrent, enabling positive, negative, or near-zero output states depending on the programming voltage.

Gate-controlled modulation of photocurrent magnitude and polarity underlies the device's fundamental operating principle. The programmable multi-level response facilitates structured analog-domain manipulation of the sensed signal, supporting compatibility with mixed-signal circuit integration and crossbar-based array configurations. The quantitative characteristics of this programmable behavior are determined through TCAD simulations [63, 64] and are presented in the following section.

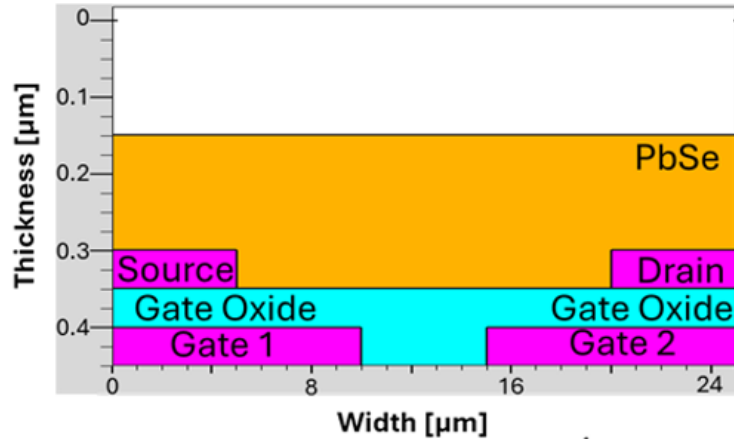


Figure 2.1 Cross-section view of the field-effect gated PbSe photodiode.

2.1.1 TCAD Simulation Setup

Device characteristics for compact model development were derived from Technology Computer-Aided Design (TCAD) simulations using the Silvaco ATLAS simulation framework. The simulated structure represents a bottom-gated field-effect PbSe photodiode architecture with independently controlled n-gate and p-gate terminals located beneath the PbSe absorber layer. In the simulation environment, the source terminal was held at ground potential, and the drain current was measured as the output. Electrostatic modulation was implemented through differential gate biasing. The n-gate voltage was swept from -0.98 V to 0 V, and the p-gate voltage from 0 V to $+0.98$ V, establishing a symmetric programming range across the device.

Mid-wave infrared illumination was applied with intensity varied from 0 to 1 mW/cm² (0 – 0.001 W/cm²). For each combination of gate bias and illumination, the steady-state drain current was recorded, generating a bias- and illumination-dependent dataset. The compact model developed in this chapter is calibrated to the TCAD dataset within the defined gate-bias programming range and operating configuration. This calibrated representation forms the device-level foundation for later circuit-level and crossbar-based integration.

2.2 Photocurrent Characteristics Analysis

The TCAD-generated dataset demonstrates a coupled dependence of drain current on both programming voltage and incident illumination intensity within the defined operating range. Variation of the differential gate bias across the programming window, spanning -0.98 V to $+0.98$ V for n-gate and p-gate voltages, results in a corresponding increase in drain current magnitude. Elevated gate bias levels enhance the internal electric field in the PbSe layer, which promotes carrier separation and increases photocurrent.

Within the examined programming window, the drain current exhibits an approximately linear dependence on gate bias at a fixed illumination level. Likewise, at a constant programming voltage, the photocurrent increases linearly with incident MWIR intensity over the simulated range of 0 to 1 mW/cm². This linear relationship between photocurrent and illumination persists across the range, and the scaling behavior remains consistent for all programmed bias levels.

Collectively, these results demonstrate that the drain current exhibits a structured and monotonic dependence on both gate bias and illumination within the examined domain. This predictable behavior facilitates the development of a compact behavioral model, which is presented in the subsequent section.

2.3 Compact Model Formulation

The compact model of the programmable PbSe photodiode was implemented as a behavioral current source within the HSPICE simulation environment. The objective of the model is to capture the coupled dependence of drain current on programming voltage and illumination intensity using a computationally efficient analytical representation suitable for crossbar-level simulation.

The device contains two electrostatic gate terminals, denoted as g_p and g_n . The effective programming voltage is defined as the differential gate bias:

$$V_{diff} = V(g_p, g_n) = V_{gp} - V_{gn} \quad (2.1)$$

The differential quantity determines the electrostatic band modulation within the PbSe layer. By adjusting the relative gate biases, the device transitions between different electrostatically formed junction configurations, resulting in programmable photocurrent magnitude and polarity.

In the circuit-level abstraction, illumination intensity is introduced as an independent control input represented by an electrical node l . The light-dependent

scaling factor is defined as:

$$V_{light} = V(l, 0) \quad (2.2)$$

where $V(l, 0)$ represents the voltage at the illumination node relative to ground. This variable serves as a proportional representation of incident MWIR intensity, enabling controlled scaling of the photocurrent within the SPICE environment.

Following the definition of the programming and illumination control variables, the drain photocurrent is expressed as a calibrated analytical function of V_{diff} and V_{light} . Analysis of the TCAD dataset indicates that the drain current increases with programming voltage and illumination intensity within the defined operating range. The illumination dependence is modeled as a proportional scaling term, while the voltage dependence is captured using a polynomial fitting formulation to reproduce the extracted device characteristics accurately. The photocurrent is therefore expressed as:

$$I_{ph} = (a_0 + a_1 V_{diff} + a_2 V_{diff}^2) V_{diff} V_{light} \quad (2.3)$$

where a_0 , a_1 , and a_2 are fitting coefficients obtained through calibration against the TCAD-generated drain current data. This formulation preserves the experimentally observed increase in current with both programming voltage magnitude and illumination intensity, while maintaining a compact structure suitable for large-scale circuit simulation. An initial co-efficient extraction was performed using polynomial regression to match the TCAD current-voltage data across the operating region. Subsequent refinement of the fitting coefficients within the research group improved the agreement between the compact model and the TCAD dataset, and the calibrated coefficient set is adopted for all simulations presented in this thesis.

The behavioral formulation in (2.3) was implemented in HSPICE using a voltage-controlled current source between drain and source terminals. A high-value

resistor $R_{dc} = 1 \text{ T}\Omega$ is included between drain and source to provide a DC return path and prevent floating nodes during operating-point computation. Due to its large resistance value, this element does not influence the modeled photocurrent under normal bias conditions but ensures numerical robustness in mixed-signal simulations. This compact representation enables efficient integration of the programmable PbSe device into crossbar-based sensing architectures and subsequent analog front-end circuits, while preserving consistency with the underlying TCAD-validated device behavior.

2.4 Model Calibration and Validation

The coefficients of the compact model were derived using polynomial regression applied to drain current data generated by Technology Computer-Aided Design (TCAD) simulations. The TCAD dataset consisted of discrete drain current values measured as functions of programming voltage and illumination intensity within the specified operating region. For coefficient extraction, the TCAD current data were analyzed as a function of programming voltage and illumination. Since the photocurrent increases proportionally with illumination intensity within the operating range, the voltage-dependent behavior was first identified by observing how the current varies with V_{diff} . A polynomial function of V_{diff} was then fitted to the TCAD data to capture this voltage dependence, while illumination was modeled as a multiplicative term that scales the overall current magnitude.

Following the definition of the programming and illumination control variables, a second-order polynomial function of V_{diff} was selected to represent the voltage-dependent conductance component:

$$G(V_{diff}) = a_0 + a_1 V_{diff} + a_2 V_{diff}^2 \quad (2.4)$$

The fitting coefficients a_0 , a_1 , and a_2 were determined by minimizing the deviation between the polynomial model and the corresponding TCAD current values across the defined bias range. Standard least-squares regression was used to obtain the optimal set of coefficients.

2.4.1 Quantitative Validation and Error Metrics

To evaluate the accuracy of the compact behavioral model, the modeled drain current was compared directly against the TCAD-generated dataset across the defined programming voltage and illumination range. A total of 4,900 operating points were evaluated.

The primary accuracy metric used in this validation is the relative error, defined as:

$$Relative\ Error = \frac{I_{TCAD} - I_{model}}{I_{TCAD}} \quad (2.5)$$

The accuracy of the compact model was evaluated by comparing the model drain current against the TCAD dataset across 4,900 operating points. The resulting error statistics are summarized in Table 2.1.

Table 2.1 Compact Model Validation Metrics

Metric	Value	Percentage/ Units
Total points compared	4900	-
Mean relative error	0.0063252	0.63%
Median relative error	0.0062302	0.62%
Maximum relative error	0.013764	1.37%
Maximum absolute current error	1.3552×10^{-9}	A

Statistical results presented in Table 2.1 indicate strong agreement between the compact behavioral model and the TCAD dataset throughout the evaluated

operating region. The mean relative error is below 1%, and the maximum deviation does not exceed 1.4% across 4,900 data points. The maximum absolute current deviation is confined to the nanoampere range. These findings confirm that the polynomial-based behavioral formulation accurately represents programmable photocurrent characteristics within the modeled voltage and illumination window.

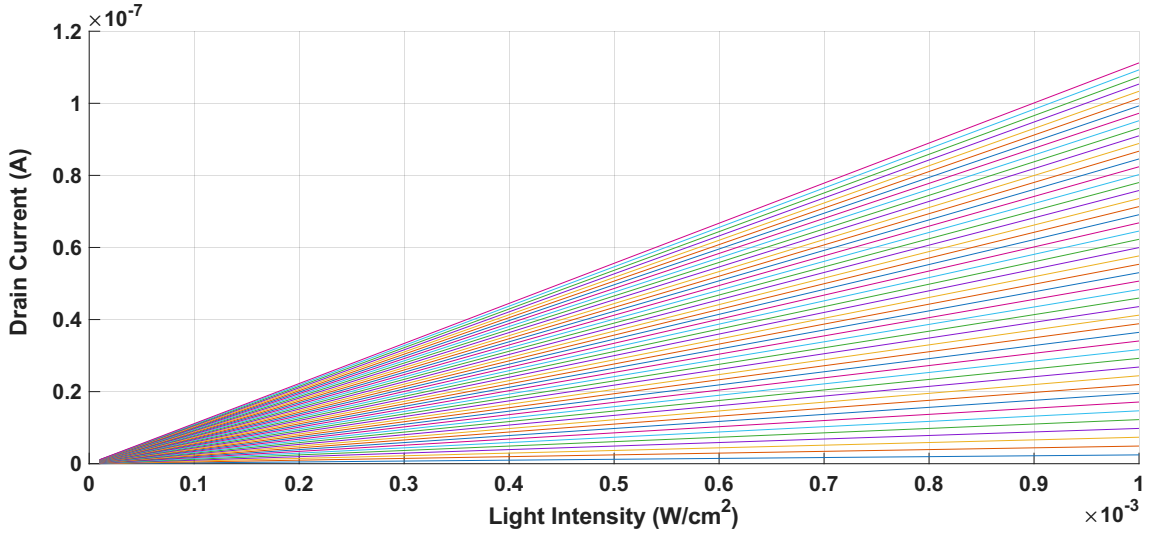


Figure 2.2 Drain current as a function of illumination intensity for multiple programming voltages.

Figure 2.2 presents the modeled drain current as a function of illumination intensity for various programming voltages. The drain current increases monotonically with both illumination intensity and programming bias throughout the evaluated operating range. At each fixed bias level, a near-linear relationship between drain current and illumination intensity is observed, and the current magnitude rises with increasing programming voltage. These results demonstrate that the compact model accurately captures the voltage-dependent programmability and illumination-dependent scaling characteristics identified in the TCAD data.

CHAPTER 3

SYSTEM ARCHITECTURE AND DESIGN

This chapter presents the system-level architecture of the mixed-signal sensing framework developed in this thesis. Building upon the compact behavioral model of the field-effect gated PbSe photodiode established in Chapter 2, the circuit-level organization of the crossbar array and its associated readout chain are described. The chapter begins with an overview of the Retinomorphic In-Sensor Processing Engine, the broader system context in which this work is situated, followed by a detailed description of the 4×4 PbSe crossbar array and the programmable gate biasing network implemented using 4-bit R-2R digital-to-analog converters.

3.1 Retinomorphic In-Sensor Processing Engine

The Retinomorphic In-Sensor Processing Engine (RISPE) was proposed in [RISPE paper] as a cross-layer hardware-software co-design framework for energy-efficient mid-wave infrared machine vision. The architecture is described briefly here to establish the system context within which the analog readout chain developed in this thesis operates. Unlike conventional frame-based imaging systems that separate sensing from processing, RISPE integrates programmable photocurrent generation and analog computation at the pixel level, significantly reducing data movement and power consumption. The architecture exploits the programmable bipolar photocurrent response of field-effect gated PbSe photodiodes, where gate-controlled electrostatic doping enables multi-level analog outputs that directly encode neural network weights for first-layer inference.

3.1.1 System Organization

At the system level, RISPE consists of multiple processing units, each implemented as an $N \times N$ retinomorphic sensor sub-array. The processing units operate concurrently and share centralized activation units and memory resources. The overall execution follows a two-phase model: a configuration phase in which digital weight parameters are converted to analog gate bias voltages and applied to the sensor array, and an inference phase in which incident MWIR radiation modulates pixel photocurrents to perform analog MAC operations through column-wise current summation. A high-level overview of the RISPE architecture is shown in Figure 3.1. The analog readout chain developed in this thesis, comprising the crossbar interface, sample-and-hold circuit, and flash ADC, forms the signal acquisition backbone of each processing unit, bridging the analog photocurrent domain and the digital output required for subsequent processing.

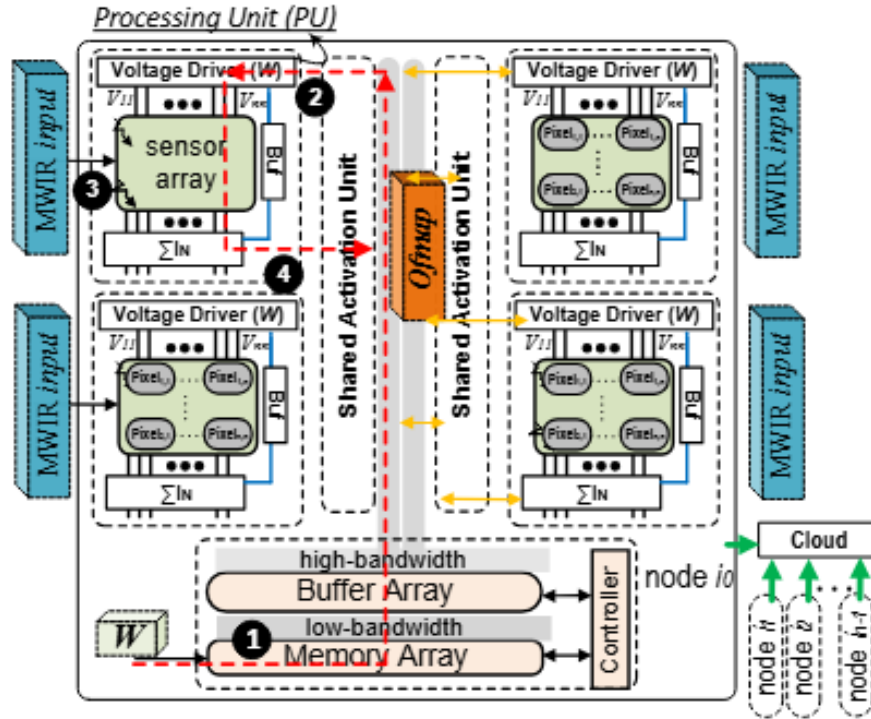


Figure 3.1 Retinomorphic In-Sensor Processing Engine architecture.

3.2 Crossbar Array and Analog Readout Architecture

The circuit-level implementation presented centers on a 4×4 PbSe photodetector crossbar array integrated with a complete analog readout chain, all of which are implemented in 180 nm CMOS technology. The overall architecture is illustrated in Figure 3.2. The system includes four R-2R digital-to-analog converters (DACs) that generate programmable row bias voltages, a 4×4 crossbar array of PbSe compact model devices, a passive transimpedance stage for current-to-voltage conversion, a transmission-gate sample-and-hold circuit, a two-stage RC anti-aliasing filter, and a 4-bit flash analog-to-digital converter (ADC) for digitization. Each column of the crossbar functions as an independent readout channel, with photocurrents from all four row devices accumulating at the shared column node before processing by the analog front-end.

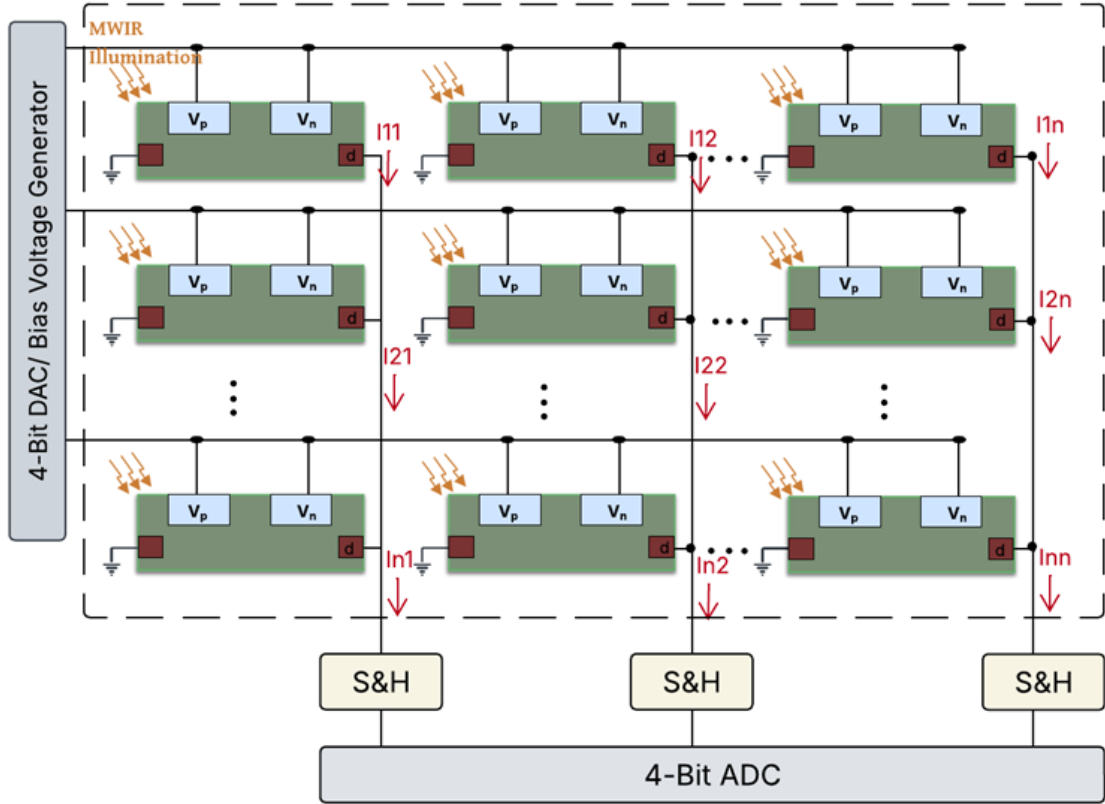


Figure 3.2 $N \times N$ PbSe crossbar array with integrated analog readout chain.

3.2.1 Crossbar Topology

The 4×4 crossbar is organized such that all devices within a given row share a common gate bias line, and all devices within a given column share a common drain output node. The source terminal of each PbSe device is connected to ground, establishing a common reference for photocurrent measurement. Under MWIR illumination, each device generates a drain photocurrent determined by its programmed gate bias and incident light intensity, as governed by the compact model formulation presented in Chapter 2. These individual photocurrents accumulate passively at the column output node according to Kirchhoff's current law, producing a total column current that represents the aggregate response of all four row devices:

$$I_{total,j} = \sum_{i=1}^N I_{ij} \quad (3.1)$$

where I_{ij} denotes the drain photocurrent of the device at row i and column j . This passive summation mechanism enables analog MAC computation without additional active circuitry, inheriting the energy efficiency of the RISPE architecture at the circuit level.

3.3 Programmable Gate Bias Generation

Programmable gate biasing of the crossbar array is achieved through four independent 4-bit R-2R ladder digital-to-analog converters, one assigned to each row. The R-2R architecture was selected for its inherent linearity [65], minimal component count, and straightforward implementation in 180 nm CMOS technology. Each DAC accepts a 4-bit binary input and produces a corresponding analog output voltage within a designated non-overlapping range, as summarized in Table 3.1.

Table 3.1 R-2R 4-BIT DAC Row Voltage Ranges

Row	Voltage Range (V)
Row 1	0 – 0.35
Row 2	0.40 – 0.60
Row 3	0.65 – 0.82
Row 4	0.85 – 0.98

The non-overlapping voltage assignment ensures that each row operates under a distinct programming regime, enabling independent photocurrent control across all rows simultaneously without inter-row interference.

3.3.1 Dual-Gate Biasing

Each PbSe device in the crossbar array is controlled by two gate terminals: a p-gate and an n-gate. The positive DAC output voltage V_{gp} is applied directly to the p-gate terminal of all devices in row i , while a voltage-controlled source generates the corresponding negative gate voltage such that $V_{gn} = -V_{gp}$. This differential biasing establishes an effective programming voltage across each device as defined in Equation 2.1, directly governing the photocurrent magnitude generated by each device under a given illumination level, as described by the compact model in Equation 2.3. Increasing V_{diff} enhances the internal electric field within the PbSe layer, promoting more efficient carrier separation and increasing the drain photocurrent magnitude. The dual-gate differential biasing scheme therefore provides a direct and linear mechanism for programming the photocurrent response of each row independently, supporting the multi-level analog computation capability of the crossbar array.

Under the defined illumination range of $10 \mu\text{W}/\text{cm}^2$ to $1 \text{ mW}/\text{cm}^2$, the total column current at each output node ranges from approximately 140 nA to 230 nA, establishing the input signal range for the analog front-end described in Chapter 4.

CHAPTER 4

ANALOG SIGNAL CONDITIONING

The photocurrents generated by the PbSe crossbar array are weak analog signals in the nanoampere range that require precise current-to-voltage conversion, buffering, sampling, and noise filtering prior to digitization. In conventional CMOS image sensor readout chains, transimpedance amplifiers (TIAs) based on operational-amplifier feedback are commonly used for current sensing [65, 7]. However, active TIA topologies impose stability constraints and bandwidth limitations at high transimpedance values, rendering passive resistive conversion more suitable for the operating current range of this system. This chapter details the complete analog front-end signal-conditioning chain implemented in 180nm CMOS technology, including the passive resistive load, voltage buffer, transmission-gate sample-and-hold circuit, and two-stage RC filter network.

4.1 Resistive Load

Current-to-voltage conversion at each column output node employs a passive resistive load between VDD and the column node. The column photocurrent induces a voltage drop across the load resistor, reducing the node voltage from VDD and generating an output voltage inversely proportional to the total column current. This passive topology is selected over active feedback designs because the high resistance required for measurable voltage swings from nanoampere-level currents would impose excessive gain-bandwidth demands on operational amplifier-based implementations. Passive resistive conversion mitigates these challenges but produces a non-inverting signal relationship, which is compensated for by the ADC’s reference ladder.

To ensure uniform distribution of the output voltage swing across the valid input range of the sample-and-hold stage, each readout channel is assigned a column-specific

resistance. This assignment is needed because the total column photocurrent varies across columns. The variation is due to differences in illumination intensity and gate bias programming. The four load resistors are summarized in Table 4.1.

Table 4.1 Per-Column Resistance Values

Column 1	3.67M Ω	18.35 ns
Column 2	4.41 M Ω	22.05 ns
Column 3	5.30 M Ω	26.50 ns
Column 4	5.85 M Ω	29.25 ns

The resistance values are selected to ensure that the resulting column voltage swing remains within 0.77 V to 1.17 V across the full illumination range of 10 $\mu\text{W}/\text{cm}^2$ to 1 mW/cm^2 , aligning with the reference range of the flash ADC described in Chapter 5.

4.2 Voltage Buffer

After transimpedance conversion, the column voltage passes through a unity-gain voltage buffer before reaching the sample-and-hold switch. Buffering is essential because the resistive load node exhibits a very high source impedance, typically several megaohms, which would otherwise result in significant voltage droop when the sample-and-hold capacitor draws a transient charging current upon switch closure [65]. In the absence of buffering, the inrush current from the hold capacitor would discharge the resistive load node, causing a systematic sampling error proportional to the hold capacitance and the switch-on resistance. The buffer addresses this issue by providing a low-impedance replica of the column voltage at its output. A 100 Ω isolation resistor is placed in series between the buffer output and the transmission gate input to dampen high-frequency inrush transients without significantly affecting the settling time constant.

4.3 Transmission-Gate Sample-and-Hold Circuit

The sample-and-hold circuit acquires the buffered column voltage during sampling and maintains its stability throughout the ADC conversion window. Such circuits are fundamental components of analog-to-digital converters, as they prevent input changes during comparator evaluation and eliminate aperture errors. The transmission gate topology is selected over a simple NMOS switch because the complementary PMOS path compensates for the NMOS device's input-dependent on-resistance, resulting in uniform charge injection and reduced signal-dependent distortion across the input voltage range[66, 65]. The transmission gate comprises an NMOS transistor with $W/L = 20 \mu\text{m}/0.18 \mu\text{m}$ and a complementary PMOS transistor with $W/L = 40 \mu\text{m}/0.18 \mu\text{m}$. The PMOS width is set to twice that of the NMOS to compensate for the two-fold difference in carrier mobility between holes and electrons in 180 nm CMOS technology, thereby equalizing the on-resistance contributions of both devices. The hold capacitor is 5 pF, which balances charge storage adequacy with settling time. The worst-case settling time constant occurs at Column 4: $\tau = 5.85 \text{ M}\Omega \times 5 \text{ pF} = 29.25 \text{ ns}$, which is within the 45 ns evaluation window allocated before `clk_cmp` rises.

4.4 Two-Stage RC Anti-Aliasing Filter

Comparator switching in flash ADCs generates kickback noise, which is injected back through the comparator input capacitance onto the signal node and can potentially corrupt the held voltage on the sampling capacitor if not adequately suppressed. To mitigate this effect, a two-stage RC filter network is inserted between the sample-and-hold output and the ADC input. Two-stage filter topologies are preferred over single-stage designs in high-speed ADC interfaces because they provide steeper attenuation of kickback transients while maintaining a flat passband response within the signal bandwidth.

The first stage comprises a 1 k Ω series resistor and 1 pF shunt capacitor, forming a single-pole low-pass filter with a cutoff frequency of:

$$f_{c1} = 1/(2\pi \times 1\text{k}\Omega \times 1\text{pF}) \approx 159 \text{ MHz} \quad (4.1)$$

The second stage sits directly at the ADC input node. It uses a 2 k Ω series resistor and a 2 pF shunt capacitor. This yields:

$$f_{c2} = 1/(2\pi \times 2\text{k}\Omega \times 2\text{pF}) \approx 40 \text{ MHz} \quad (4.2)$$

The cascaded configuration progressively attenuates kickback transients while allowing the baseband signal to pass without distortion. A 50 M Ω bleeder resistor at each filter node prevents floating and suppresses DC offset accumulation at the ADC input.

4.5 Timing and Signal Integrity

The complete analog front-end operates on a 100 ns clock period. The timing sequence is allocated to ensure that each stage fully settles before the subsequent stage is activated, in accordance with established design principles for pipelined analog readout chains. The clock signal CLK rises at 10 ns and falls at 20 ns, defining a 10 ns sampling window during which the transmission gate is closed and the hold capacitor charges to the input voltage. The hold phase begins at 20 ns when CLK falls and the switch opens. The ADC comparison clock clk_cmp rises at 45 ns, providing 25 ns of hold settling time after switch opening, which is sufficient for the two-stage RC filter to attenuate any switching transients. The latch clock clk_lat rises at 85 ns, capturing the fully resolved thermometer output into the D-latches. The total conversion latency from the end of sampling to digital output capture is 65 ns, resulting in a throughput of 10 MS/s per column and 40 MS/s aggregate across all four columns.

CHAPTER 5

MULTI-BIT FLASH ANALOG-TO-DIGITAL CONVERTER DESIGN

The analog front-end described in Chapter 4 delivers a conditioned voltage signal to the analog-to-digital converter (ADC) input at the end of each sampling cycle. This chapter details the design and transistor-level implementation of the 4-bit flash ADC used to digitize the column voltages. The flash architecture was chosen for its single-cycle conversion capability and inherently parallel operation, both of which are critical for achieving the high-throughput requirements of the column-parallel readout chain [65]. The ADC is implemented entirely in 180 nm CMOS technology and consists of a parallel StrongARM comparator array, output buffers, D-latches, and a thermometer-to-binary encoder.

5.1 Flash ADC Architecture

A flash ADC converts an analog input to a digital code by simultaneously comparing the input against all $2^N - 1$ reference levels within a single clock cycle, producing a thermometer-coded intermediate result that is subsequently encoded into binary [65]. This parallel comparison eliminates the multi-cycle latency found in successive-approximation or pipeline architectures, making the flash topology well suited to the low-latency, high-throughput requirements of the crossbar readout system. For 4-bit resolution, 15 parallel comparators are required, each evaluating the input against a unique reference voltage during a single comparison clock cycle.

5.1.1 Reference Voltage Ladder

The 15 reference voltages are uniformly distributed across the valid input range of 0.52 V to 1.30 V, covering the full output swing of the analog front-end described in Chapter 4. The reference step size is approximately 55.7 mV per least significant bit

(LSB), with reference voltages assigned from $V_{ref1} = 0.520$ V to $V_{ref15} = 1.300$ V in uniform increments. This reference range is precisely matched to the resistive load output swing, ensuring that all valid input levels correspond to unique digital codes without saturation or missing codes.

5.2 StrongARM Comparator

Each of the 15 comparators is implemented using a StrongARM latch topology, which enables high-speed dynamic comparison with near-zero static power dissipation [65]. The StrongARM comparator operates in two phases controlled by the comparison clock (`clk_cmp`). During the precharge phase, when `clk_cmp` is low, the PMOS precharge devices reset both output nodes to V_{DD} . When `clk_cmp` transitions high, the tail NMOS device activates and the differential input pair steers current asymmetrically according to the voltage difference between the input and the assigned reference. The cross-coupled PMOS load devices provide positive feedback, rapidly amplifying the initial imbalance into a full rail-to-rail digital output. The transistor sizing is as follows: tail device $W/L = 10 \mu\text{m}/0.18 \mu\text{m}$, differential input pair $W/L = 6 \mu\text{m}/0.18 \mu\text{m}$ each, cross-coupled PMOS loads $W/L = 8 \mu\text{m}/0.18 \mu\text{m}$ each, and precharge devices $W/L = 4 \mu\text{m}/0.18 \mu\text{m}$ each. This sizing balances regeneration speed with input-referred offset and kickback noise.

5.3 Output Buffering and Latching

The raw output of each StrongARM comparator is buffered by a two-inverter chain before being delivered to a level-sensitive D-latch. The buffer restores full logic levels and isolates the dynamic comparator node from the capacitive load of the latch input, preventing charge sharing that could impair regeneration speed [cite Razavi]. Each D-latch is transparent during the high phase of the latch clock (`clk_lat`) and holds its output when `clk_lat` is low. The two-phase clocking scheme separates comparator

evaluation from output capture: `clk_cmp` rises at 45 ns to initiate evaluation, and `clk_lat` rises at 85 ns to latch the settled thermometer output, providing 40 ns of evaluation time before capture.

5.4 Thermometer-to-Binary Encoder

The 15 latched thermometer bits, C1 through C15, are converted to a 4-bit binary output (b3, b2, b1, b0) using a combinational edge-detection encoder constructed entirely from standard CMOS logic gates. The encoder first computes 15 edge terms, D1 through D15, where $D_k = C_k \text{ AND } (\text{NOT } C_{k+1})$ identifies the thermometer transition boundary, and $D_{15} = C_{15}$ is directly buffered. The four binary bits are then generated as OR combinations of the relevant edge terms:

- $b_3 = \text{OR}(D_8\text{--}D_{15})$
- $b_2 = \text{OR}(D_4\text{--}D_7, D_{12}\text{--}D_{15})$
- $b_1 = \text{OR}(D_2, D_3, D_6, D_7, D_{10}, D_{11}, D_{14}, D_{15})$
- $b_0 = \text{OR}(D_1, D_3, D_5, D_7, D_9, D_{11}, D_{13}, D_{15})$

Each OR tree is implemented as a chain of 2-input CMOS OR gates followed by an output buffer, resulting in minimal propagation delay beyond the comparator regeneration time.

5.5 Conversion Timing and Throughput

The complete ADC conversion pipeline operates within the 100 ns system clock period. During the precharge phase, all comparator outputs are reset to VDD. At 45 ns, `clk_cmp` rises and all 15 comparators evaluate simultaneously. At 85 ns, `clk_lat` captures the resolved thermometer code into the D-latches, and the encoder immediately generates the 4-bit binary output. The total conversion latency from the

end of sampling to digital output is 65 ns per cycle, resulting in a throughput of 10 MS/s per column and 40 MS/s aggregate across all four columns. SPICE simulations confirm that the ADC accurately tracks time-varying input voltages and updates the 4-bit digital output synchronously every 100 ns, thereby validating the timing closure of the complete conversion pipeline.

CHAPTER 6

SYSTEM-LEVEL INTEGRATION AND VALIDATION

Previous chapters described the individual circuit blocks that comprise the mixed-signal readout chain. This chapter presents integrated system-level simulation results, obtained by connecting all blocks within a unified HSPICE simulation environment. End-to-end operation from optical input to digital output is demonstrated through transient simulation, and the power consumption of the complete system is analyzed.

6.1 End-to-End System Integration

The complete system integrates a 4×4 PbSe crossbar array, four R-2R DACs, load resistors, voltage buffers, transmission-gate sample-and-hold circuits, two-stage RC filters, and four parallel 4-bit flash ADCs within a single HSPICE netlist. Each of the four columns functions as an independent readout channel. Photocurrents from all four row devices accumulate at the shared column node before processing by the analog front-end and digitization by the ADC. All simulations utilize the 180 nm CMOS technology library, a supply voltage of 1.8 V, and a 100 ns system clock period.

6.2 Device-Level Validation: Compact Model Response

Prior to full system integration, the programmable photocurrent response of the compact PbSe device model was validated under staircase gate-bias and pulsed-illumination conditions. Figure 6.1 presents the transient photocurrent response of a single PbSe device under stepwise-programmed gate bias and pulsed MWIR illumination. The bottom panel shows the gate voltage $V_{gp,gn}$ increasing in discrete staircase steps, with each step representing a distinct programming level applied to the device gates. The top panel shows the corresponding photocurrent response,

where, at each gate bias level, multiple illumination pulses of varying intensity are applied, producing current pulses of proportional amplitudes. As the gate voltage increases across successive steps, the peak photocurrent at each illumination level increases accordingly, demonstrating that the device gain is directly controlled by the gate bias. This behavior confirms two key characteristics of the compact model: the linear scaling of photocurrent with illumination intensity at a fixed bias level, and the bias-dependent amplification of the photocurrent response across the full programmable range. Both properties are essential for reliable analog MAC computation within the crossbar array, where each device must faithfully encode the product of its programmed weight and the incident light intensity.

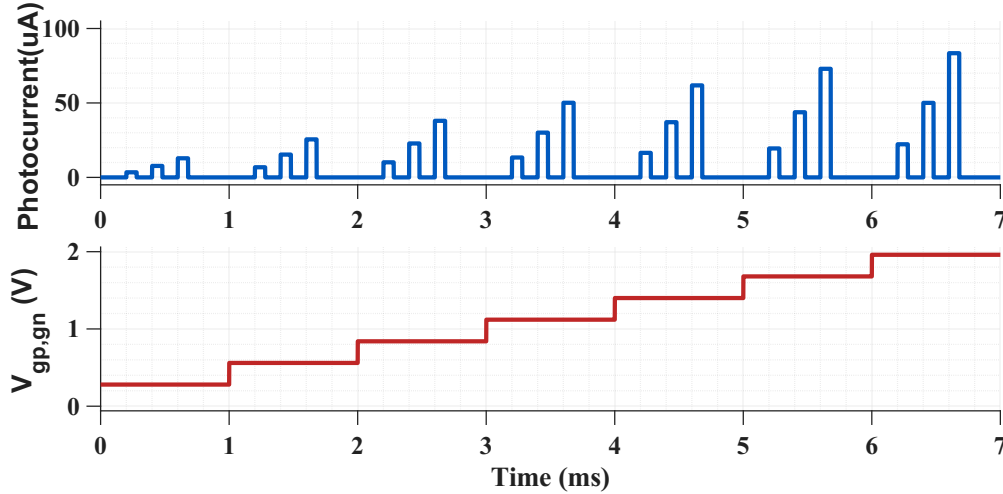


Figure 6.1 Drain current response under stepwise-programmed voltage and pulsed illumination.

6.3 Full System Transient Simulation

Figure 6.2 presents the transient simulation results for the complete integrated system. The figure displays, from top to bottom: the 10 MHz system clock; the four DAC row voltages (V_{ROW1} – V_{ROW4}) operating within their respective non-overlapping ranges; the MWIR illumination intensity patterns applied to Column 1 devices; the individual photocurrents (I_{11} – I_{41}) and total column current (I_{total}) at the Column 1 output node;

the resistive load output voltage (V_{out}) after current-to-voltage conversion; and the 4-bit ADC digital outputs (b_3 – b_0) tracking the analog input.

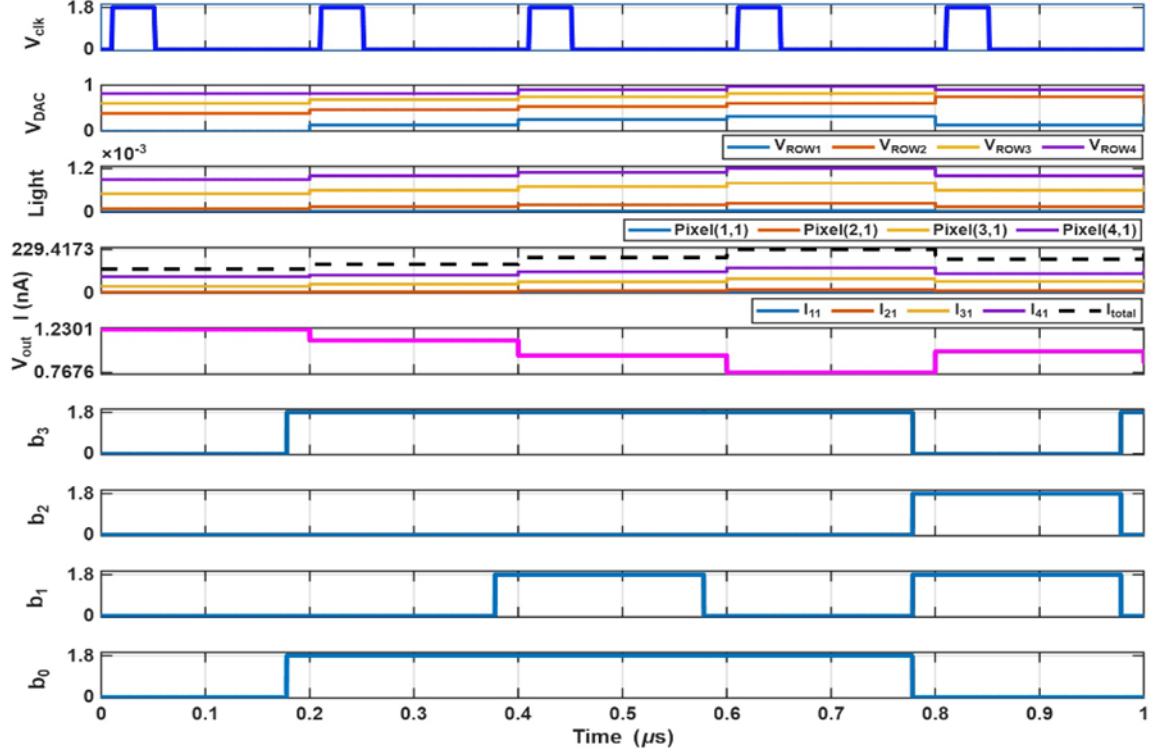


Figure 6.2 Transient simulation of the 4×4 PbSe crossbar array with integrated analog front-end.

The simulation results confirm correct end-to-end operation across all circuit blocks. The DAC voltages transition between programmed levels as the 4-bit input codes change, resulting in corresponding changes in the photocurrent generated by each row device. The total column current ranges from approximately 140 nA to 230 nA across the full illumination range of $10 \mu\text{W}/\text{cm}^2$ to $1 \text{ mW}/\text{cm}^2$, consistent with the crossbar current analysis in Chapter 3. The resistive load output voltage responds inversely to the column current, varying between 0.77 V and 1.17 V. The ADC digital outputs update synchronously every 100 ns, accurately encoding the analog voltage into 4-bit binary codes with a conversion latency of 65 ns per cycle.

6.4 Power Consumption Analysis

The power consumption of each subsystem was measured from the integrated HSPICE simulation using average current measurements over the steady-state operating window. These results are summarized in Table 6.1.

Table 6.1 Power Consumption Summary

Block	Power
DAC Switching	702 nW
PbSe Crossbar Array	651.18 nW
Analog Front-End (Resistive load + S/H + RC)	1.16 μ W
Flash ADC (4 columns)	52.65 mW
Total	$\sim$ 52.65 mW

The flash ADC accounts for over 99% of the total system power consumption, primarily due to the dynamic switching activity of 60 StrongARM comparators operating at 10 MHz across four parallel columns. In comparison, the analog front-end, crossbar, and DAC contribute negligibly, reflecting the energy efficiency of the passive TIA topology and the low operating currents of the PbSe crossbar devices. The energy per conversion is 2.98 nJ/sample for the ADC, while the DAC switching energy per cycle is 42.1 fJ, confirming the low overhead of the programmable biasing network relative to the digitization stage.

6.5 System-Level Scalability Analysis

An architecture-level evaluation was conducted to assess the scalability of the proposed readout architecture within the complete RISPE system. This evaluation considered a 640×512 array configuration across four weight precision settings: 1-bit, 2-bit, 4-bit, and 8-bit, denoted as [A:1-bit] through [A:8-bit]. The analysis included five components: weight programming (WeightApply), ADC conversion,

sample-and-hold, row decoding, and miscellaneous overhead. The results are summarized in Table 6.2.

Table 6.2 RISPE System-Level Energy and Execution Time Breakdown (640×512 Array)

Configuration	Energy/Frame	Time/Frame	FPS	Avg Power
	(μJ)	(μs)		(mW)
[A:1-bit]	1.528	219.44	4557	6.96
[A:2-bit]	1.531	219.44	4557	6.98
[A:4-bit]	1.538	219.44	4557	7.01
[A:8-bit]	1.552	219.44	4557	7.07

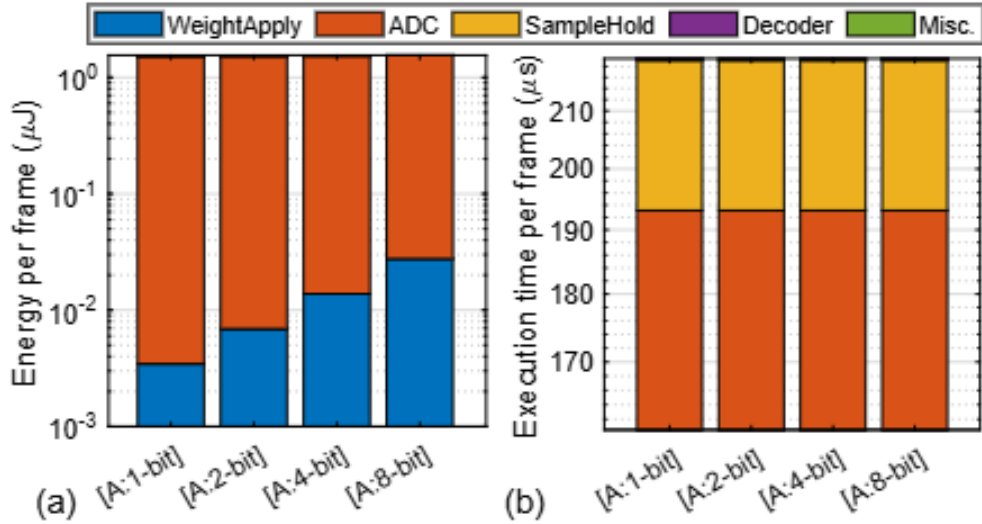


Figure 6.3 Per-frame breakdown of (a) energy consumption and (b) execution time across ADC bit precision configurations.

Figure 6.3 illustrates the energy and execution time breakdown across the four precision configurations. The ADC accounts for over 99% of the total frame energy in all configurations. WeightApply energy increases linearly with bit precision, from 3.45 nJ at 1-bit to 27.58 nJ at 8-bit, but remains negligible compared to the ADC's 1.524 μJ per frame. The sample-and-hold and decoder components together contribute less

than 0.003% of total frame energy, demonstrating the energy efficiency of the passive analog front-end design.

The total frame latency remains constant at 219.44 μs across all precision configurations, yielding a throughput of over 4,557 FPS regardless of bit width. This invariance arises because frame latency is dominated by the WeightApply operation, contributing approximately 160.56 μs due to its per-pixel invocation across the full array. The ADC and S&H contribute 32.56 μs and 25.97 μs , respectively, owing to their column-level parallelism. These results confirm that increasing ADC bit precision does not reduce system throughput, as the performance bottleneck lies in weight programming rather than digitization precision.

6.6 Comparison with State-of-the-Art MWIR Imagers

Table 6.3 compares the proposed RISPE architecture against commercial MWIR imaging platforms [67, 68, 69].

Table 6.3 Comparison of RISPE With Commercial MWIR Imaging Platforms

Metric	RISPE	[68]	[69]	[70]
Resolution	640×512	640×512	640×480	640×512
Detector	PbSe (uncooled MWIR)	MWIR FPA	HgCdTe	T2SL
Cooling	None	Cryogenic	Cryogenic	Cryogenic
Total Power	7.01 mW	≈ 4 W	≈ 5 W	≈ 4.5 W
Frame Rate (FPS)	4557	60	60	30
Latency / Frame	219 μs	16.7 ms	16.7 ms	33.3 ms
Energy / Frame	1.538 μJ	~ 66 mJ	~ 83 mJ	~ 150 mJ
Energy / Pixel	4.69 pJ	~ 200 nJ	~ 270 nJ	~ 460 nJ
ADC Resolution	4-bit	12–16 bit	14 bit	16 bit
Data Movement / Frame	$\ll 0.1$ MB	~ 0.6 –1.3 MB	~ 0.6 –1.3 MB	~ 0.6 –1.3 MB
Data Movement / Pixel	≈ 1 –4 bit	12–16 bit	14 bit	16 bit
Compute Capability	In-sensor MAC	None	None	None
TOPS/W	~ 0.1 (est.)	≈ 0	≈ 0	≈ 0

The proposed RISPE architecture demonstrates substantial advantages over conventional MWIR imaging systems across all key performance metrics. Operating at only 7.01 mW without cryogenic cooling, RISPE achieves over 4,557 FPS with a frame latency of 219 μ s, compared to 30–60 FPS and millisecond-scale latency in commercial platforms [67, 68, 69]. The energy per frame is reduced by approximately four to five orders of magnitude relative to conventional systems, which consume 66–150 mJ per frame. By performing first-layer MAC operations directly within the sensor array, effective data movement is reduced from 0.6–1.3 MB per frame in traditional pipelines to sub-0.1 MB, significantly reducing downstream bandwidth and memory requirements. Unlike conventional systems that rely on high-resolution 12–16 bit full-frame ADC conversion, RISPE employs 4-bit column-parallel conversion tailored for feature extraction, enabling a fundamentally different trade-off between energy efficiency, throughput, and inference accuracy.

CHAPTER 7

CONCLUSION

This thesis presents the circuit-level modeling and mixed-signal integration of a field-effect gated PbSe photodiode crossbar architecture for mid-wave infrared retinomorphic sensing, contributing to the broader development of the Retinomorphic In-Sensor Processing Engine (RISPE) for energy-efficient MWIR machine vision. Conventional MWIR imaging systems incur significant power and bandwidth overhead due to the separation of sensing and processing functions. The proposed architecture addresses this limitation by enabling programmable analog-domain computation directly at the sensor interface, reducing data movement and supporting in-sensor multiply-and-accumulate operations for neural network inference.

At the device level, a TCAD-calibrated compact behavioral model of the field-effect gated PbSe photodiode was developed in HSPICE, achieving a mean relative error of 0.63% across 4,900 operating points. Building upon this model, a 4×4 PbSe crossbar array with four independent 4-bit R-2R DACs was designed to enable programmable dual-gate biasing across non-overlapping row voltage ranges. A complete analog signal conditioning chain implemented in 180 nm CMOS comprising a passive resistive load, transmission-gate sample-and-hold circuit, and two-stage RC anti-aliasing filter conditions the accumulated column photocurrents prior to digitization. A 4-bit flash ADC with 15 StrongARM comparators achieves 40 MS/s aggregate throughput with a conversion latency of 63.6 ns. Full system integration in HSPICE validated correct end-to-end operation from optical input to digital output across the full illumination range of $10 \mu\text{W}/\text{cm}^2$ to $1 \text{ mW}/\text{cm}^2$, with total system power of 52.65 mW dominated by the ADC. Architecture-level evaluation across a 640×512 array demonstrates over 4,557 FPS throughput at 7.01 mW, confirming

system scalability across ADC bit precision configurations. Compared to commercial MWIR imaging platforms operating at 30–60FPS with power budgets of 4–5W and requiring cryogenic cooling, RISPE achieves over four to five orders of magnitude reduction in energy per frame while eliminating the need for cooling entirely. These results establish a circuit-level framework for scalable MWIR retinomorphic sensing within the RISPE architecture.

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